Nozzle user manual and integration guide

Operator gestures, displays, witness case and Fabien’s camera-based integration.

Illustrated edition 1 ·5 November 2026. This manual describes the **existing Nozzle module, now with MWB and PA32**. Welded plate P03 is a different application; do not directly transfer its axes, PA controls or settings.

> Open byte-ndt-nozzle-formation.streamlit.app, then **3 · Manual scan → Load backscatter witness test — MWB45°/4MHz**. Export session data before loading this preset.

Purpose: practise gestures and explain responses from a reduced model. Teaching modes, scalar direct field and computed backscatter are distinguished explicitly. This is not a qualified industrial inspection procedure or an approved certification examination.

Contents

1. 1. Four tabs and two control levels
2. 2. Specimen and Nozzle coordinates
3. 3. MWB, access and flaw setup
4. 4. Select the correct field model
5. 5. PA32: sector, incident field and backscatter
6. 6. Probe gestures and stops
7. 7. Interpret the operator views
8. 8. Repeatable scalar witness exercise
9. 9. Record, export and prepare an examination
10. 10. Colours, gain and differences from plate P03
11. 11. Fabien’s camera handoff — no headset
12. 12. Troubleshooting
13. 13. Scope and reviewer feedback

1. Four tabs and two control levels

Tab	Use
1 · Nozzle 3D	Understand geometry, walls and intersections. Rotating the camera does not move the probe.
2 · Specimen setup	Select MWB, access zones, scan plan and flaws. Planned points are not acquired measurements.
3 · Manual scan	Move/orient the probe, select field model, adjust gates and record points.
4 · User guide	Download FR/EN PDFs and find notes, protocols, geometry exports and glossary.

The sidebar controls language, purpose, specimen dimensions, material and preparation assumptions. The operator dashboard in tab 3 has its own motion, gate, gain and display controls.

Export before changing Streamlit settings: language changes now retain this browser tab's scan session. Geometry, method or scenario changes use a separate session; export before changing specimens. New PDF downloads do not rerun the script, but keep a working export.

The supplied offline package starts with LANCER_NOZZLE.bat on Windows beside app.py. First installation may need Internet. PDF/HTML manuals can be read without running Python.

2. Specimen and Nozzle coordinates

The model represents a set-through branch in a main pipe. Witness geometry: main Ø400×20mm, nozzle Ø160×12mm, axis angle 63°, offset 0, main length 800mm, external nozzle length 320mm. Bevel 35°, gap 2mm, land 2mm, nominal weld width 14mm and reinforcement 3mm are adjustable assumptions.

Quantity	Convention
World frame	Right-handed, millimetres. Origin at main-pipe centre; main axis is **Z**.
Azimuth	Angle around Z:0° towards +X,90° towards +Y. It is not a distance in millimetres.
Z position	Distance along the main pipe; negative values are valid. An 800mm pipe ends at−400/+400mm.
Depth	Radial distance from outer wall inward, distinct from global Z.
Local skew	Tangent-plane probe heading from +Z, distinct from position azimuth and MWB refracted angle.

In the geometry view, red is the outer-surface intersection and turquoise the nominal geometric root at the main inner wall. Dashed preparation-edge curves are illustrative offsets, not measured fusion boundaries.

The nozzle is visually trimmed, not a verified Boolean CAD weld solid. Faceted root/cap profiles are repeatable assumptions. Microstructure parameters are metadata only; no anisotropic or grain-scattering calculation is performed.

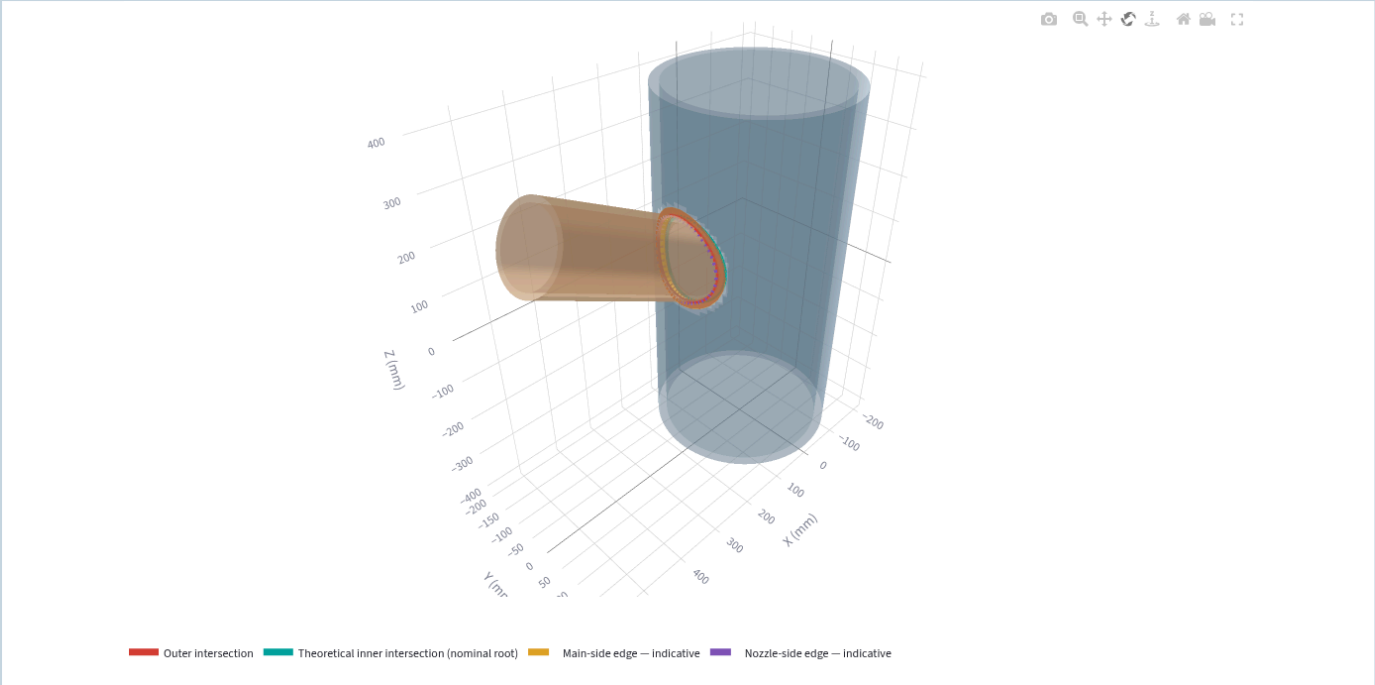


Fig. 1 — Reference geometry: main pipe and 63° nozzle, intersection curves and approximate profiles.

3. MWB, access and flaw setup

Nozzle offers **single-element MWB** (45/60/70°, proposed 2/4/5MHz) or **32-element 5MHz PA**. Actual probe/wedge reference and dimensions remain pending.PA uses the plate's initial array parameters, with fields and projections adapted to the Nozzle cylindrical frame.

The equivalent field uses entered VT. Carbon-steel witness values are VL5920m/s, VT3240m/s and density 7850kg/m³. Wedge transfer, refraction and curved contact are not solved by the scalar model.

In Training, **Main-wall scan demo** supplies five artificial wall targets for gesture practice, not real weld-bevel flaws. **Custom specimen** uses the project register. Incomplete azimuth/Z/depth/orientation prevents a flaw from generating a teaching response.

The proposed register has two 6×3mm lack-of-fusion surfaces, one 2mm-diameter inclusion and two 4×2mm cracks; dimensions are editable. Their position and normal are scenario data, not automatic weld-face placement. The witness enables MF1 only.

The access table records angular/Z bounds, available clearance, wedge footprint, travel and margins. **PLANNED means planned, not inspected.** Unassessed areas allow teaching exploration, not proven real access. An exclusion without usable bounds blocks travel.

4. Select the correct field model

Field choice	Calculated / illustrated behaviour	Associated A-scan
Computed backscattered field — scalar	Finite MWB aperture and finite reflecting face/illuminated sphere; phase, directivity and complex aperture reception. Fixed 0.05 reference,−60..0dB display. Direct illumination, single scattering.	Envelope from computed reception and centre-path time. No added heuristic root/cap echoes.

Field choice	Calculated / illustrated behaviour	Associated A-scan
Reflected field — first teaching model	Teaching energy along rays with illustrative spreading/reflection weights, not an elastic solver.	Uncalibrated heuristic geometry/flaw responses. Magenta return is an explanatory path.
Computed 3D volume / Computed cut	Direct scalar Green aperture field with phase/frequency; grid-specific normalization, stopping at the first interface for the displayed field.	Remains heuristic, not reception computed from that direct field.

The greyed **Future modules — Kirchhoff/Born/GTD** menu does not select active calculations. Available scalar surface scattering is chosen under **Computed backscattered field**; this does not constitute a complete elastic Kirchhoff/Born/GTD engine.

Computed scattering is explanatory: pink surface and field remain visible even without Instructor enabled. It is not a blind exercise. Use the teaching mode and hide revealing aids for student gesture exercises.

5. PA32: sector, incident field and backscatter

Tab 2 Method selects PA32; tab 3's **Load PA32-element/5MHz witness test** supplies a repeatable case. It sets mechanical step 2mm, separate from angular step 2°. The array has 32 elements,0.5mm pitch and 12mm elevation. Element width 0.45mm and 30mm focus are demonstration values. Nominal 55°T wedge is a starting reference; wedge propagation is not calculated.

Element phases excite the finite flaw surface, and reception uses the same coherent aperture with reciprocal phases. Surface field and A-scan come from this scalar calculation, not a relabelled MWB reflected teaching field.

1. Load the PA witness and wait.
2. Check Az−50°, Z−100mm, heading 0°, selected 45°, sector 35–70° and Δθ2°.
3. GateA0–20µs, threshold 0.001, gain 10dB0dB. Direct MF1 time remains near 8.73µs; amplitude before gain is approximately0.00538 at this pose.
4. Inspect the calculated PA incident field/white ray, backscattered field and selected-angle A-scan. Change selected angle to 65°, wait and compare reception; restore 45°.
5. S-scan backprojects sector responses into Z/radial depth, direct paths only. Circumferential travel may be compressed by this projection; consult 3D.
6. Record point at several poses. B-scan uses first ray 35°, not the selected A-scan angle. Export scan and scene to retain sector angles/responses.

Real examination preparation supports PA settings/strategy, but no instrument is connected and no synthetic echo is presented as a real acquisition. Coverage, hardware wedge delay laws and acceptance thresholds are not qualified.

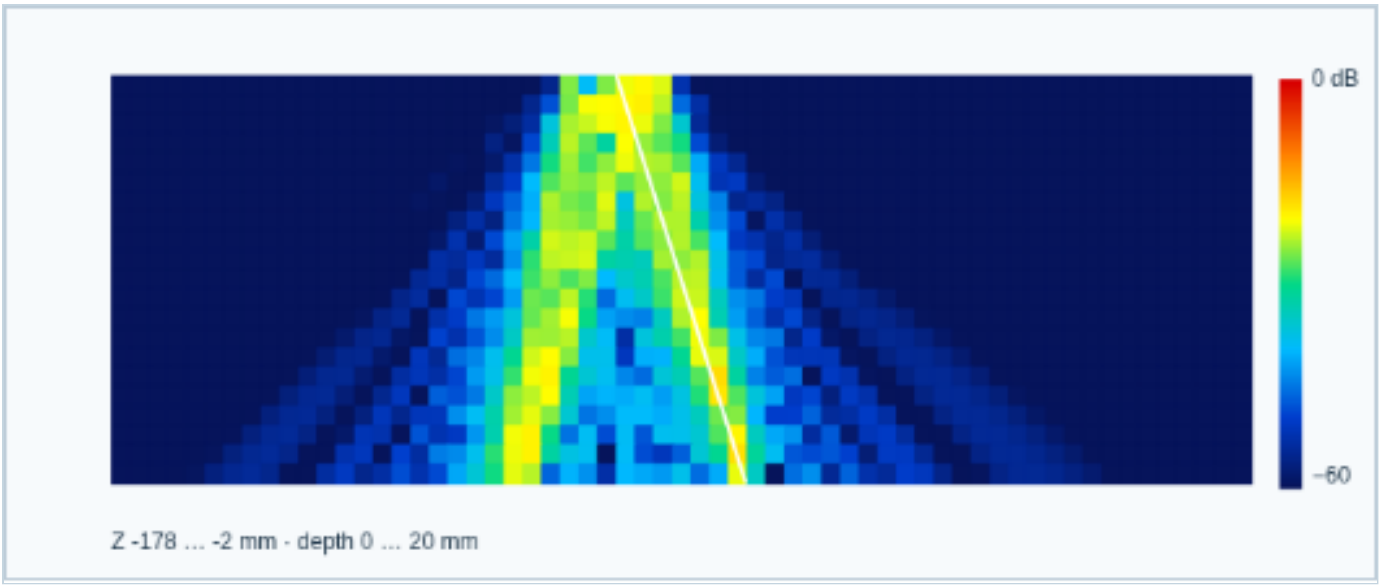


Fig. 2 — PA: phased incident field and direct ray in the cylindrical wall.

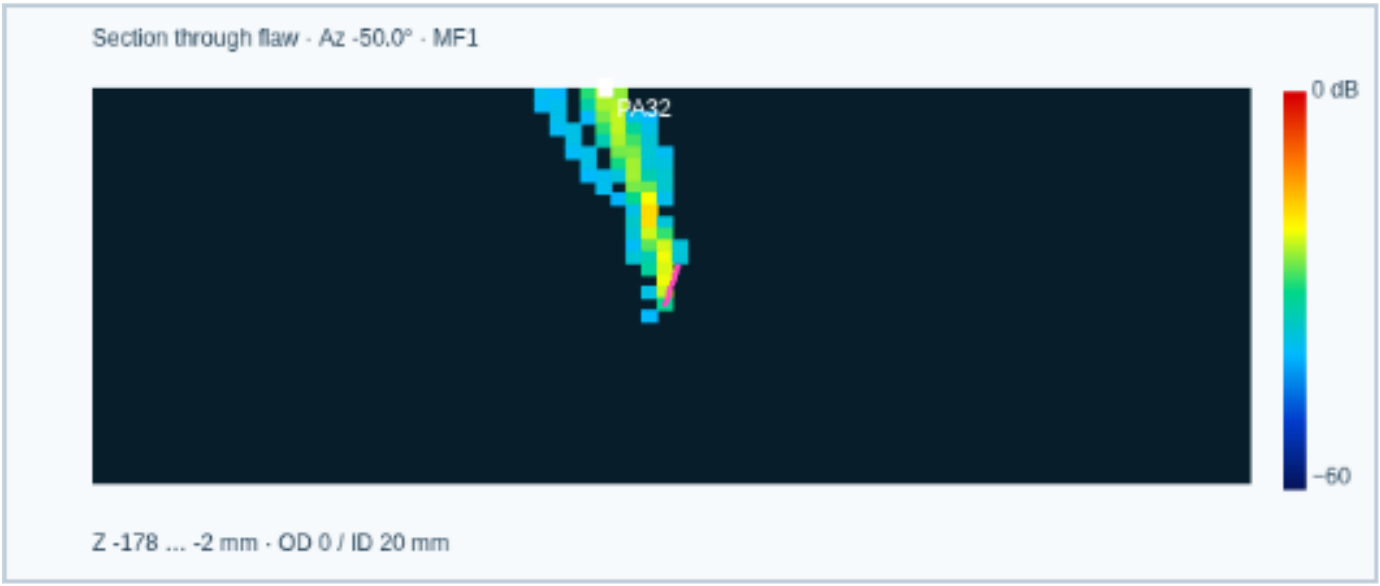


Fig. 3 — PA: scattering from the same surface, retaining phase through reception.

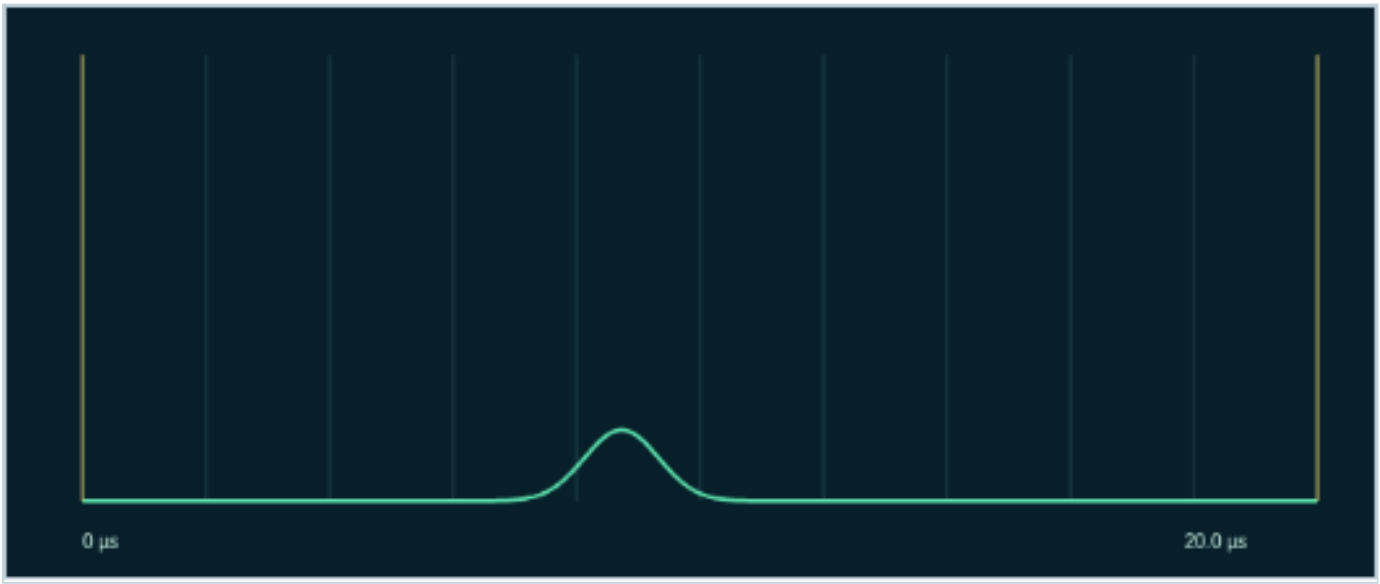


Fig. 4 — PA: selected 45° A-scan,30dB gain and 0–20µs gate.

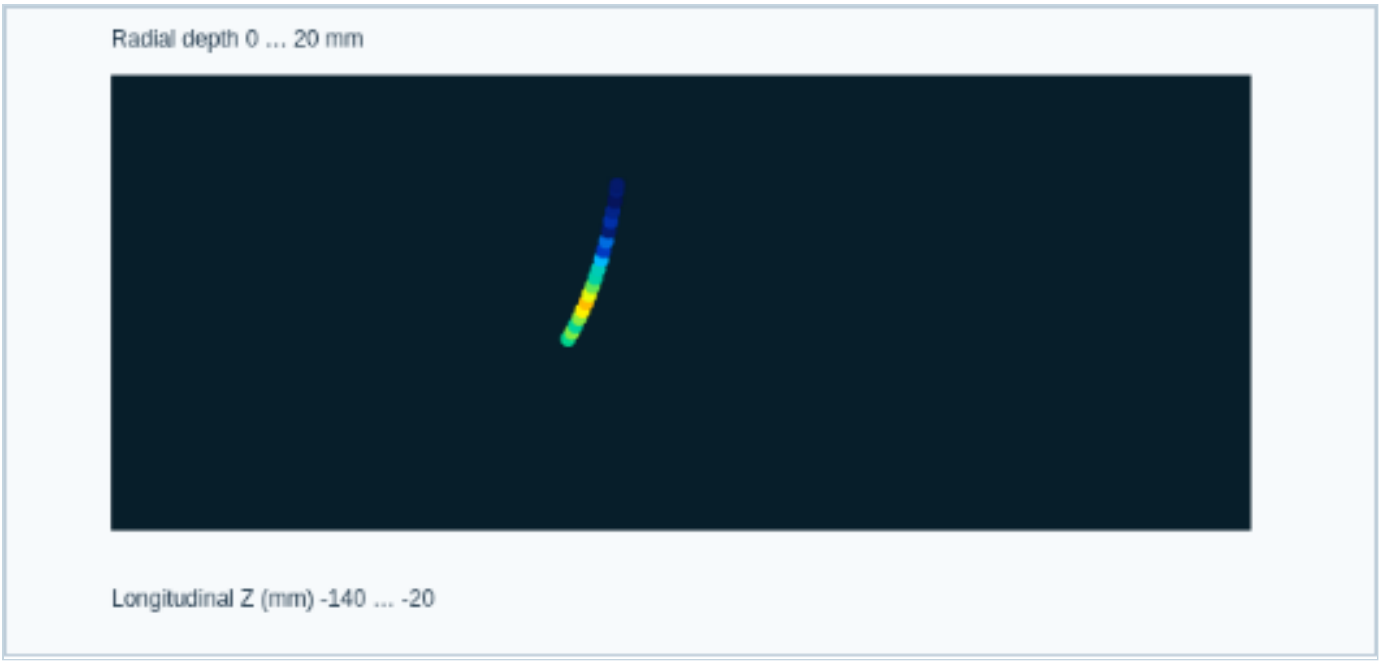


Fig. 5 — PA:35–70° backprojected Z/depth sector, not tomography.



Fig. 6 — PA: B-scan of recorded samples, first 35° ray.

6. Probe gestures and stops

1. Tab 3: Surface = Nozzle 3D; Action = Move probe.
2. Grab the MWB symbol with the left mouse button and drag on the wall. Position remains constrained to the main cylinder.
3. Wheel near the symbol changes heading.−1°/+1° buttons and Local skew provide fine adjustments.
4. For camera rotation/zoom, choose Rotate specimen, then return to Move probe to scan.
5. Unwrapped surface is a secondary rectangular representation of the cylinder.

Path	Meaning
Free	Free wall motion, manual heading and raster playback.
Main pipe axis	Circle at constant Z around the main pipe, with adjustable skew. This is not axial translation along Z.
Nozzle axis	Main-wall contour around a branch-coaxial envelope; probe aims towards the weld. Stand-off expands that envelope, not an exact geodesic distance. Path−/+ advances along it.

The probe remains on the main pipe, not the branch. Scan bounds, exclusions and footprint clearance remain active; narrow bounds can prevent a full circuit.

Raster ▶️ starts/resumes the plan in Free mode. **Reverse scan** changes travel direction without rotating the probe; **Rotate probe 180°** changes heading without reversing plan order. Raster stops at the end. STOP/Pause, Space or Escape stops motion.

Stops use an approximate centred 24×14mm footprint plus margin, not complete CAD collision or qualified curved contact. The last admitted pose is retained and a reason displayed. Back away or reverse rather than forcing the obstacle.

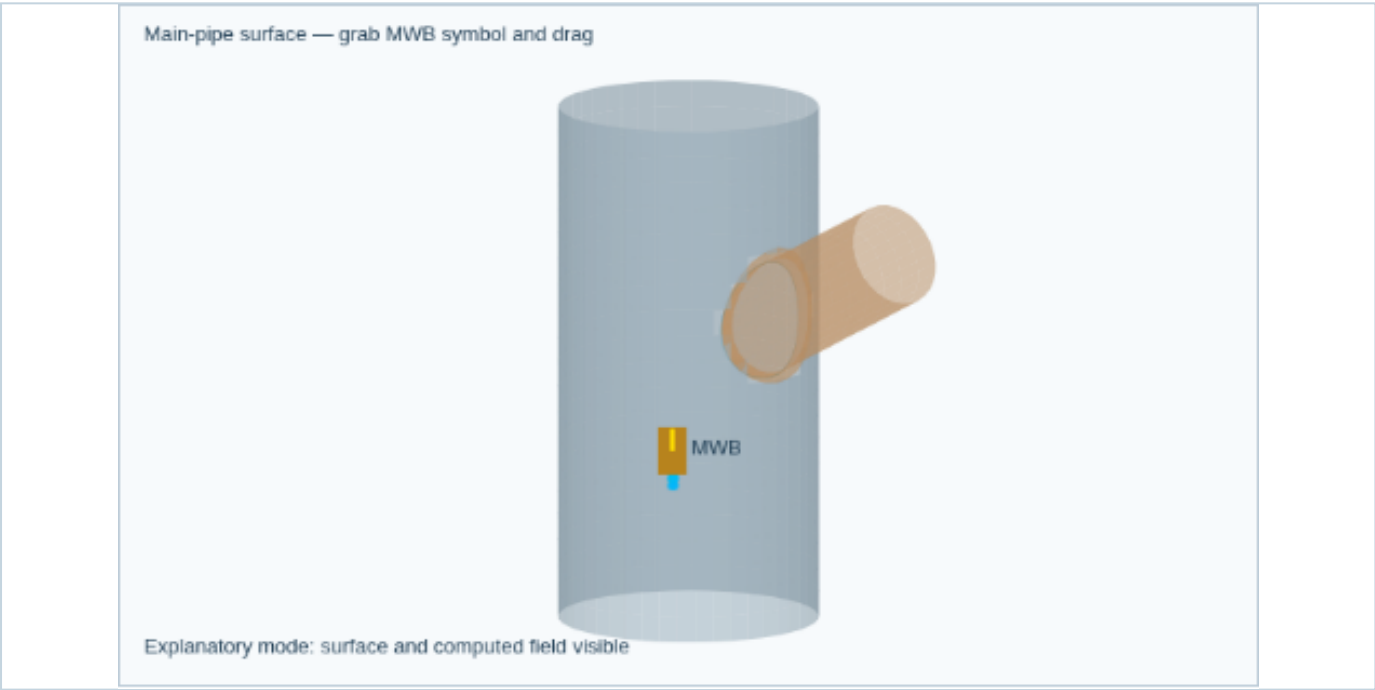


Fig. 7 — Initial MWB45°/4MHz witness pose on 3D main-pipe surface.

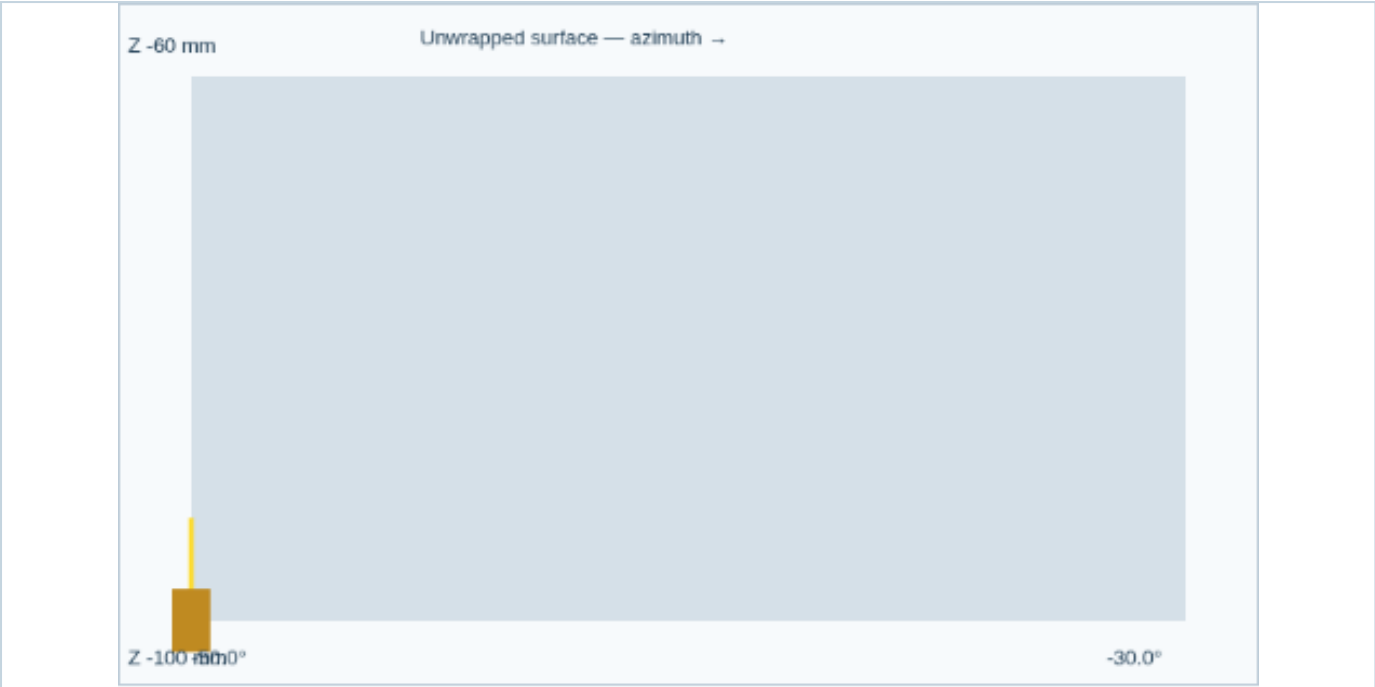


Fig. 8 — Same pose on the unwrapped wall, a secondary movement aid.

7. Interpret the operator views

Surface 3D and flaw focus

Probe symbol and rays are geometric references. In computed scattering, pink points discretize the finite flaw surface and coloured points sample its outgoing field. **The orange segment from probe to target centre is not an incident-field map.** Focus computed flaw changes the camera, not probe position.

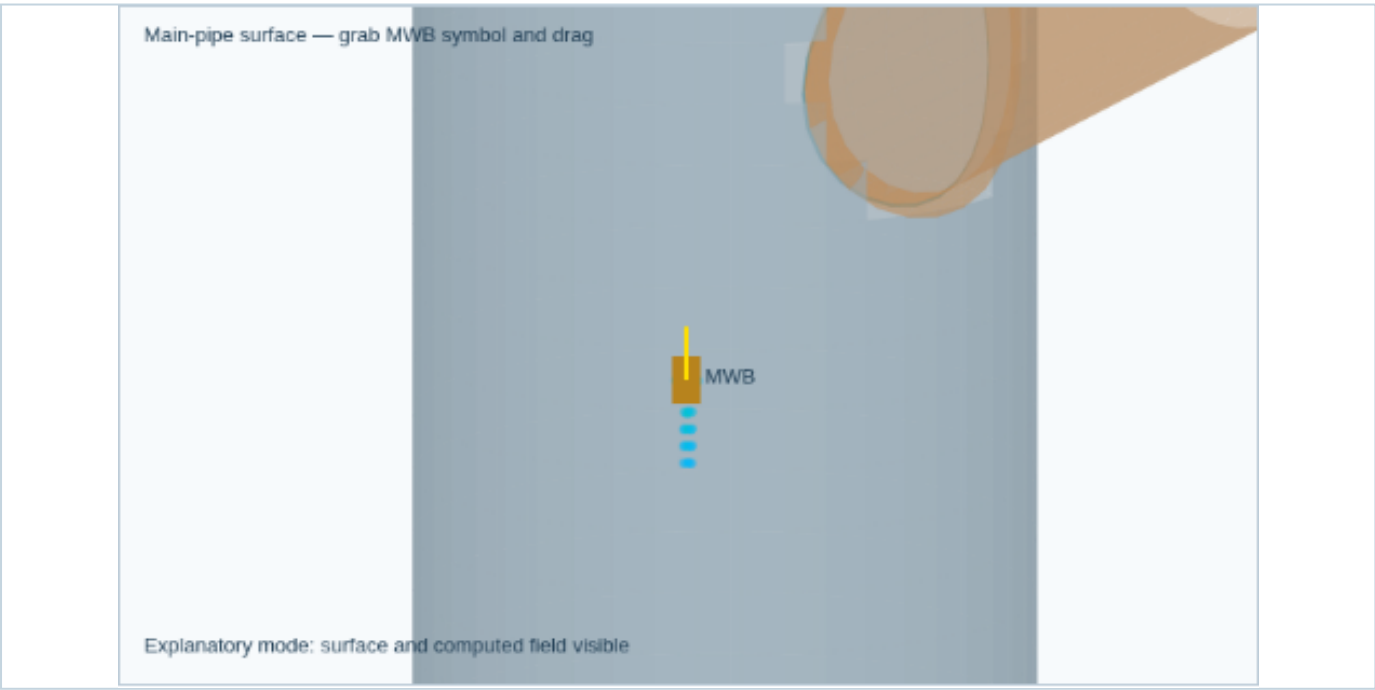


Fig. 9 — MF1 focus: finite pink surface and calculated scattering lobes. Orange line is a probe/target reference.

Section and secondary 3D view

Section is a Z/radial-depth projection; circumferential travel may look compressed. Nozzle 3D+rays clarifies it. In computed scattering, the map passes through the selected flaw: horizontal Z, vertical radial depth, azimuth stated. It does not automatically describe the entire weld volume.

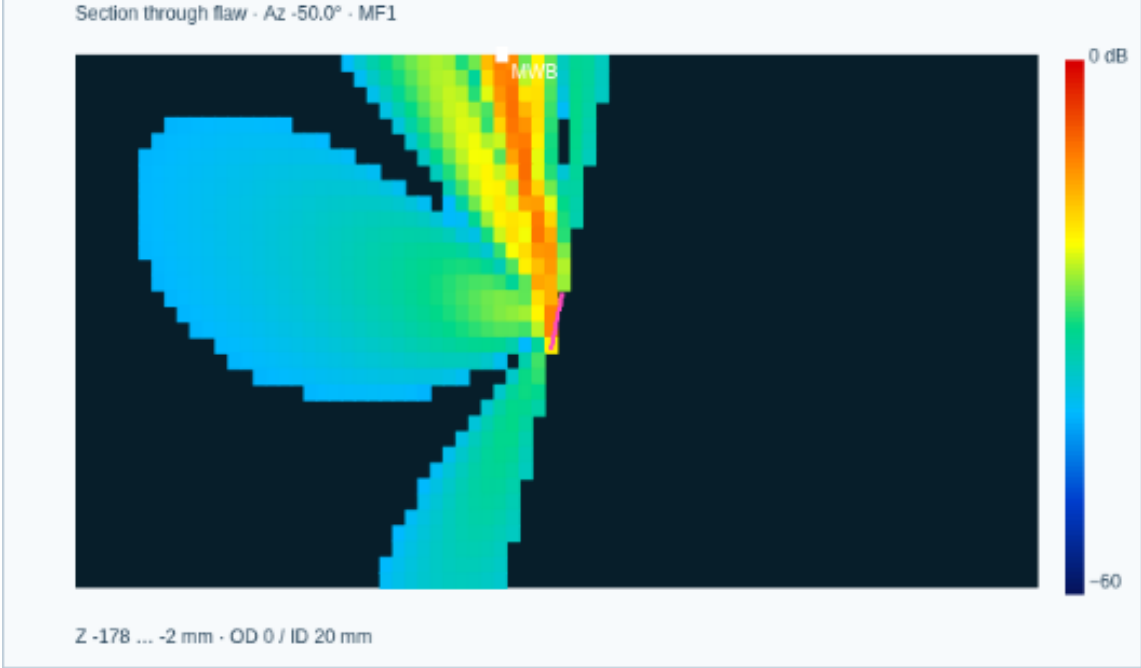


Fig. 10 — Scalar section through MF1: horizontal global Z, vertical radial depth; azimuth fixed at the flaw.

A-scan and gates

Horizontal: time in µs. Vertical: uncalibrated relative envelope. GateA/B/C selects interval and threshold. Current peak changes with pose; **stored maximum is historical**, not the current response. Gate or bounds changes reset that maximum.

In scalar mode, amplitude comes from computed reception; time is 2×distance to centre/VT. The graphic pulse is not a solved broadband impulse response. In teaching mode, geometry/flaw responses use separate heuristics.



Fig. 11 — GateA0–20µs, gain 10dB: scalar witness near 8.73µs. Inspect the current response.

Scattered field and colour cutoff

Lobes/fringes result from complex finite-surface contributions, not extra defects. Threshold(dB) hides weak points without changing computed reception. MF1 or Strongest received selects the target whose field is displayed.

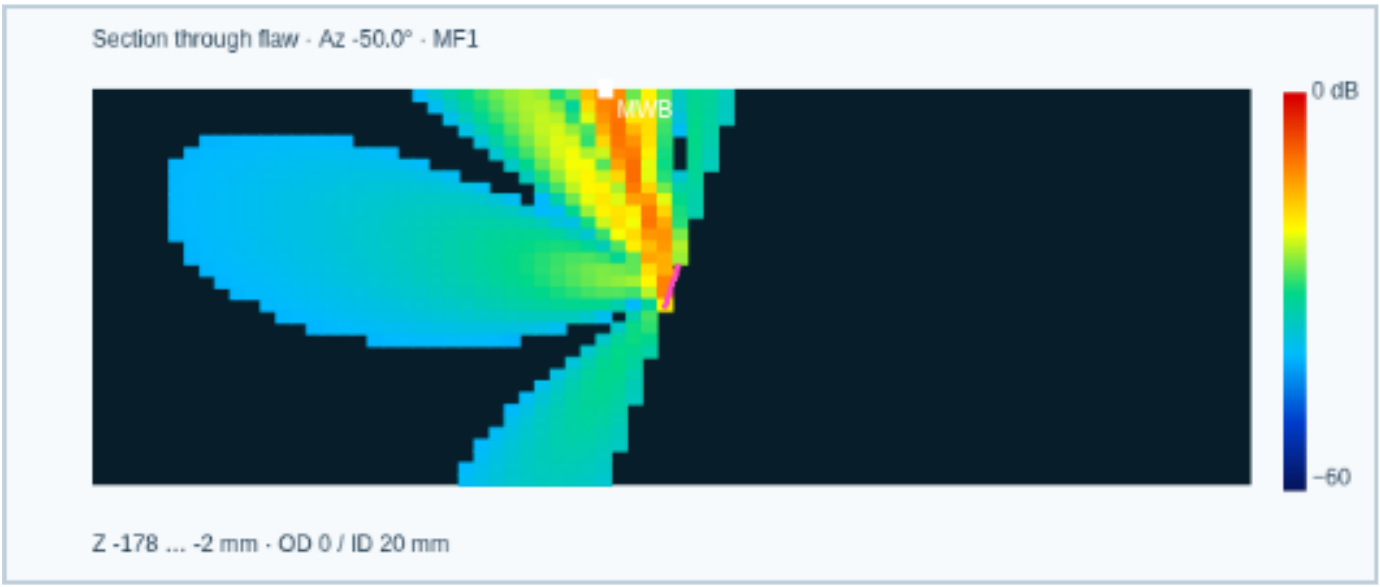


Fig. 12 — Computed backscatter, fixed 0.05 reference and−40dB display cutoff; no wall-bounce scattering.

Historical and direct-field modes

The first model helps trace reflected paths, with illustrative amplitudes. The computed direct field contains aperture interference but no solved wall-bounced field. Changing modes is not a calibrated probe comparison.

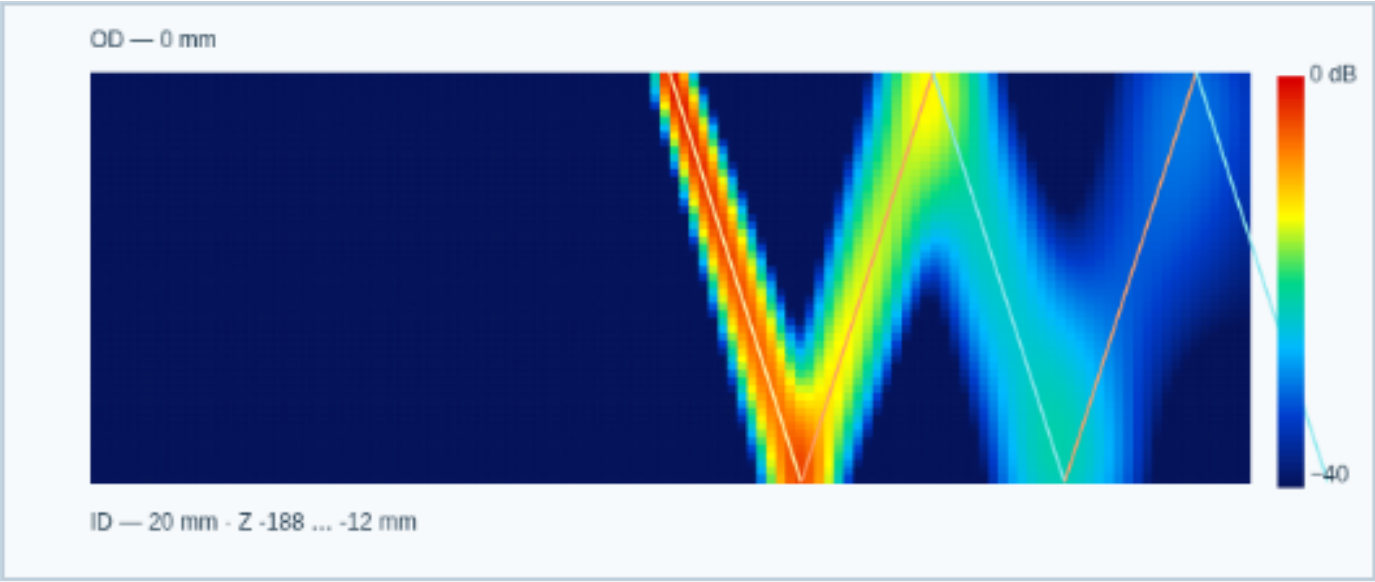


Fig. 13 — Historical reflected teaching field with heuristic amplitudes.

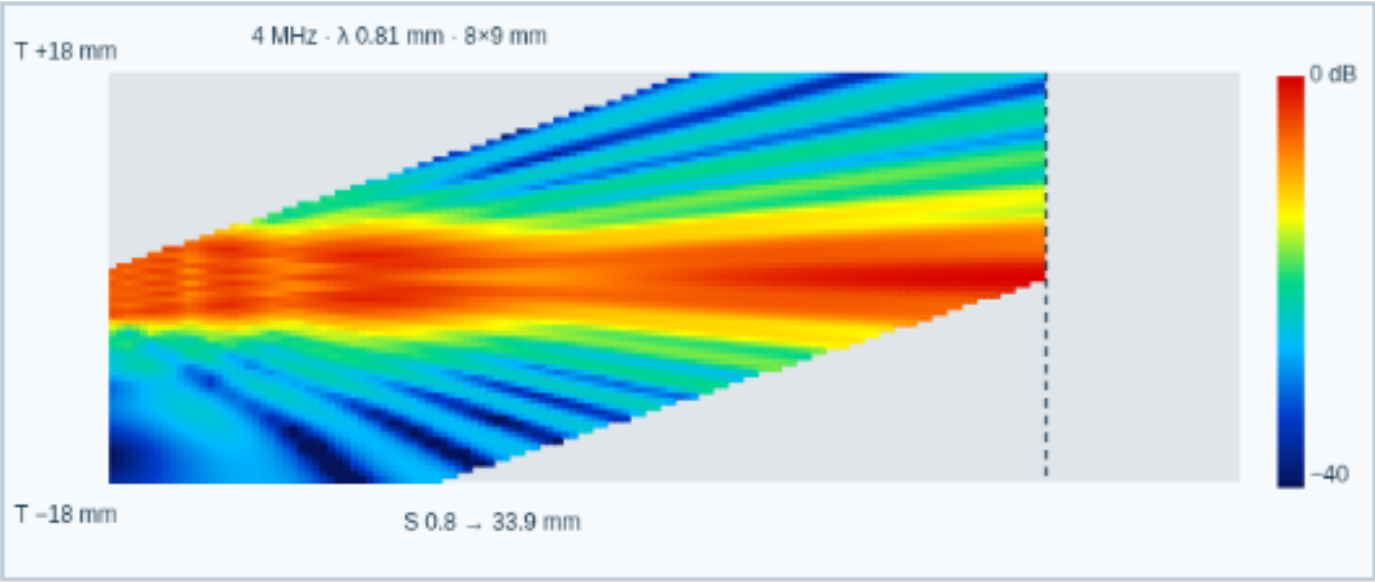


Fig. 14 — Direct scalar Green field, grid-normalized, without a computed reflected field.

8. Repeatable scalar witness exercise

1. Export personal session data, then select **Load backscatter witness test — MWB45°/4MHz**.
2. Check 400×20/160×12mm geometry,63°, VT3240m/s. Only MF1 active:6×3mm face, Az−50°, Z−90mm, depth 10mm, orientation 0°, normal tilt 45°.
3. Probe starts Az−50°, Z−100mm, heading 0°. Choose computed backscatter and MF1/Strongest received.
4. Wait for completion. Old results are withdrawn and Record point disabled while pending.
5. GateA, start 0, end 20µs, threshold 0.01, **gain 10dB**, field cutoff−40dB. Expected centre-path time is about**8.73µs**; model relative amplitude about**0.062** before display gain.
6. Focus computed flaw. Adjust heading in small increments, then compare+20° after calculation. Compare current echo, not the stored maximum.
7. Reload witness; tab 2, change MF1 normal tilt 45→80°. Return to scan and compare. Reload to restore the reference.

8. Compare 4/5MHz after reloading and wait for the new calculation. Wavelength affects phase and quadrature; this is not recolouring.
9. Change gate 0–20 to 30–40µs: direct echo leaves the gate, but harmonic field remains. Disable MF1: no flaw field/echo.

This checks the reduced model.8.73µs and 0.062 are not instrument-calibration values or weld acceptance criteria.

9. Record, export and prepare an examination

Record point stores a pose, gate, current response and geometric ray references. Motion alone builds the trail, not the same characterization samples.−6dB extent uses compatible recorded samples; it is not certified flaw size.

Export	Content / use
Scan Export JSON	NOZZLE_MANUAL_SCAN.json: configuration, trail, samples, gates, responses and stop events.
Export VR scene	NOZZLE_VR_SCENE_V1.json: frames, poses/matrices, parametric geometry, rays and active scalar field.
Test-piece JSON	Dimensions, MWB, flaws, access and metadata. Empty positions remain undefined.
Plan/access/curves/parameters CSV	Geometric preparation and traceability, not proof of UT coverage or acquisition.

Real examination preparation has no instrument acquisition connected; synthetic signals are not presented as real measurements. US319 reference/revision, cascade and criteria require documentation/validation. This mode does not award certification.

The instructor defines the scenario and hidden aids. Visible computed scattering explains responses rather than providing a blind test. Fabien’s scene JSON contains truth and must not be distributed as a secure examination.

The procedure/report panel records operator, reference and observations and exports readable HTML. Recorded samples retain dB gain, method, probe profile, model, PA angles and gates; browser timestamps do not impose physical scan speed.

10. Colours, gain and differences from plate P03

Computed scattering uses fixed reference 0.05 in model units,−60..0dB. Cutoff does not normalize the field. Direct Green field uses per-grid/frequency normalization: compare patterns, not calibrated absolute amplitude. The first teaching model is another representation.

Both methods now use dB gain: initially 10dB for MWB,30dB for PA, range 0–60dB. This matches the plate convention. Historical ×3 gain references describe earlier versions. Display gain and gates do not alter solved harmonic scattering. Colours from different modes are not interchangeable without references.

CIVA and MANTIS/GEKKO guide operator-readability review, but no exact manufacturer LUT or solver equivalence is claimed. Warm lobes mean high model amplitude, not certain flaw presence or weld rejection.

Nozzle: main axisZ, positions Az/Z; plate: weld axisX, positions X/Y. Nozzle: MWB/PA32; plate: SE/MWB/PA. Nozzle scalar scattering: direct, no wall-bounce scattering; plateP03: one opposite-face image per transmit/receive leg. Preserve these distinctions in tests and integration.

11. Fabien’s camera handoff — no headset

The goal is training controlled by PC/external camera without a headset. Nozzle currently supplies scene data and a local read API. Video detection, camera/specimen calibration and a tracked-pose input must be connected with Fabien.

```
 const api = window.BYTE_NDT;
const pose = api.getPose();
const scene = api.getScene();
const rays = api.getBeamRays();
```

These functions run inside the operator document. Streamlit hosts it in an iframe; the adapter must account for that boundary. They are not video input, network protocol or hardware commands. Do not assume the plate’s tracked-pose input API already exists in Nozzle.

rotation_matrix_local_to_world is 3×3 with basis vectors in columns: localX forward tangent, localY side, localZ outward normal. Right-handed, millimetres; multiply positions/dimensions by 0.001 for metres and transform rotations too if the engine changes coordinates.

1. Reproduce chapter 8’s witness.
2. Export scene; check emergence point, normals and scale. A camera marker centre need not be that emergence point.
3. Use pose, rays and `BYTE_NDT_MOTION_STOP` event in Fabien’s environment.
4. Check scalar_scattering.computation_pending and associated pose; never show an old field as a new pose’s result.
5. Agree tracked-pose input, confidence, tracking loss, contact, stops and accuracy/latency tests.

Outgoing field points are in scalar_scattering.field_samples_world with section/events nearby. VR_SCENE_SCHEMA.md specifies them. beam.segments_world_mm are geometric references, not computed scattered-field lines. Export is not a verified weld mesh, completed camera client or WebXR session.

12. Troubleshooting

Observation	Check
No response	Training enabled, complete active flaw position, normal, probe pose and field model. Export before restoring witness.

Observation	Check
Maximum does not decrease with rotation	Stored maximum is historical. Inspect current echo; changing gate bounds resets the maximum.
Probe does not drag	Grab its symbol, select Move probe/Free, check plan and stop reason. Rotate specimen controls camera only.
Raster unavailable	Use Free. Axis paths use dragging or Path−/+.
Magenta return disabled	Expected in computed backscatter; calculated field replaces that heuristic aid.
Field disappears after moving	Wait for the current pose; no old result should remain as a new response.
No gated signal	Event time may lie outside bounds. Gate selection does not remove harmonic field physics.
No camera picture	Nozzle does not directly read video. Connect tracking/calibration with Fabien.
Lost scan trail	Language retains the session; configuration changes use a separate session. Export before changing specimen settings.

13. Scope and reviewer feedback

Figures come from the actual local Streamlit application in both languages. Software checks cover witness, exports, languages, gestures and manual downloads, not industrial qualification.

Scattering is homogeneous, harmonic scalar, direct-illumination and single-scattering. No incident/scattered wall bounces, coupled elastic modes, wedge, curved contact, multiple interactions, GTD or full weld/branch propagation. Inclusion is an approximate reflecting sphere, not a real inclusion-contrast solver.

Useful feedback states Nozzle REV0.5 SCALAR BACKSCATTER, language, field mode, geometry, MWB/frequency, azimuth/Z/skew, flaw/normal, gate, gain, action, observed/expected result, screenshot and export. Separate gesture, pedagogy and physics comments.

Online modules: Nozzle and welded plate. Priority is Fabien’s camera handoff, followed by Jacques Marlier’s technical/teaching review; broader discussions follow.